

Galois 3

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Problem 1. Let $F = \mathbb{Q}(\omega)$ where $\omega = e^{2\pi i/3}$. Determine the splitting field of

1. $\sqrt[3]{2 + \sqrt{2}}$
2. $\sqrt{2 + \sqrt[3]{2}}$

Problem 1a.

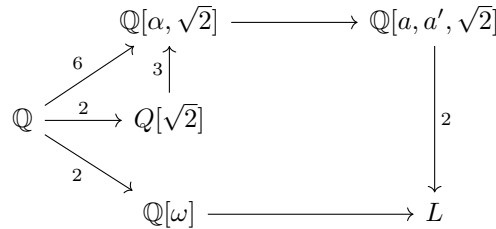
Proof. The min polynomial is $(x^3 - 2)^2 - 2 = x^6 - 4x^3 + 4 - 2 = x^6 - 4x^3 + 2$. By Eisenstein its irreducible. Let $\beta = \omega \sqrt[3]{2 + \sqrt{2}}$ and let $\gamma = \omega^2 \sqrt[3]{2 + \sqrt{2}}$, $\alpha' = \sqrt[3]{2 - \sqrt{2}}$. Define β', γ' by flipping the internal plus to a minus. The other roots of the minimal polynomial are $\alpha, \beta, \gamma, \alpha', \beta', \gamma'$. $\forall g \in \text{Gal}(L/\mathbb{Q}[\omega]), g$ fixes ω by definition of galois group, $\forall m \in L, g(\omega m) = g(\omega)g(m) = \omega g(m)$. Thus G commutes with multiplication by ω . Moreover multiplication by ω permutes the roots since $\alpha \rightarrow \beta \rightarrow \gamma \rightarrow \alpha$ and $\alpha' \rightarrow \beta' \rightarrow \gamma' \rightarrow \alpha'$.

$G \leq S_6$, permuting the 6 roots, and commutes with multiplication by ω . Multiplication by ω corresponds to 2 disjoint 3 cycles. Thus $G \leq Z(\omega)$. All disjoint 3 cycles are conjugate thus the conjugacy class of disjoint 3 cycles is

$$\frac{\frac{6*5*4}{3} * \frac{3*2*1}{3}}{2} = 20 * 2 = 40$$

By orbit stabilizer, $|Z(\omega)| = 18$. Thus $|G| = 6, 18$. Cannot be 9 because there is a subfield of degree 6, $\mathbb{Q}[\omega, \alpha]$.

$[Q[\alpha, \alpha'] : Q] = [L : Q[\omega]]$, because ω is complex and α and α' are purely real. Once we add, ω to $Q[\alpha, \alpha']$ and Q we get L and $Q[\omega]$ respectively each which is degree 2 over their smaller fields. Thus $[L : Q[\omega]]$ has the same degree as, $[Q[\alpha, \alpha'] : Q]$.



Thus if we prove that $[Q[\alpha, \alpha', \sqrt{2}] : Q] \geq 18 \implies [L : Q[\omega]] = |G| \geq 18$ and since $|G| \leq 18 \implies |G| = 18 \implies G = Z(\omega)$.

Find the minimal polynomial of α' over $Q[\alpha, \alpha', \sqrt{2}]$. $-\alpha'^3 + 2 = \alpha^3 - 2$. Thus α' is a solution to $\alpha'^3 + \alpha^3 - 4 \implies \alpha'$ is a root of $x^3 + \alpha^3 - 4$. The factors of the polynomial are $x - \alpha', x - \omega\alpha', x - \omega^2\alpha'$ and no product of linear factors is in $Q[\alpha]$. Thus the polynomial is irreducible and thus $[Q[\alpha, \alpha', \sqrt{2}] : Q[\alpha]] = 3$. $[Q[\alpha, \alpha', \sqrt{2}] : Q[\alpha]] \geq 3 \implies [Q[\alpha, \alpha', \sqrt{2}] : Q] \geq 18 \implies |G| = 18$ by the above and thus $G = Z(\omega)$. $\square$